FFT-UniBa at Cruciverb-IT: Special Length Tokens and CSP for Italian Crossword Solving

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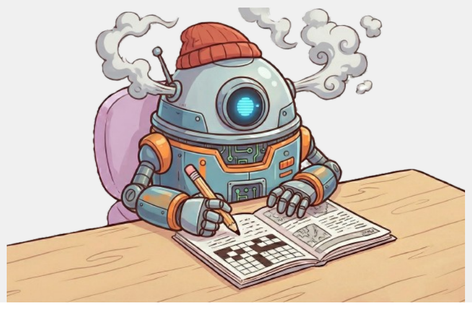
Abstract

- Two sub-tasks: answering clues and filling complete crossword grids.
- We fine-tune an Italian T5 model with special length tokens for constraint handling.
- A CSP solver with backtracking is used for grid completion.



Sub-task 1: Clue Answering

- **Goal:** Given a clue and expected length, produce ranked list of candidates.

Sub-task 2: Grid Solving

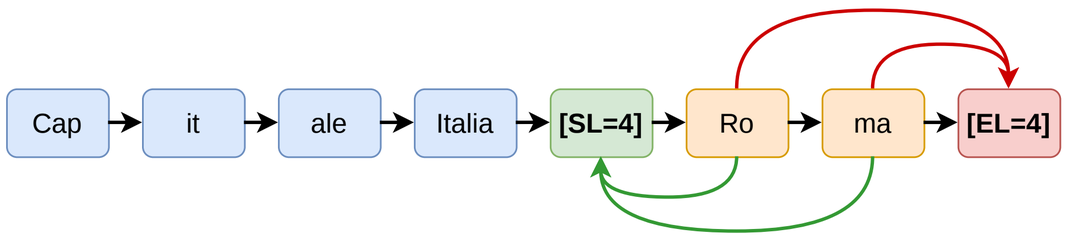
- **Goal:** Fill crossword grid satisfying all intersection constraints simultaneously.

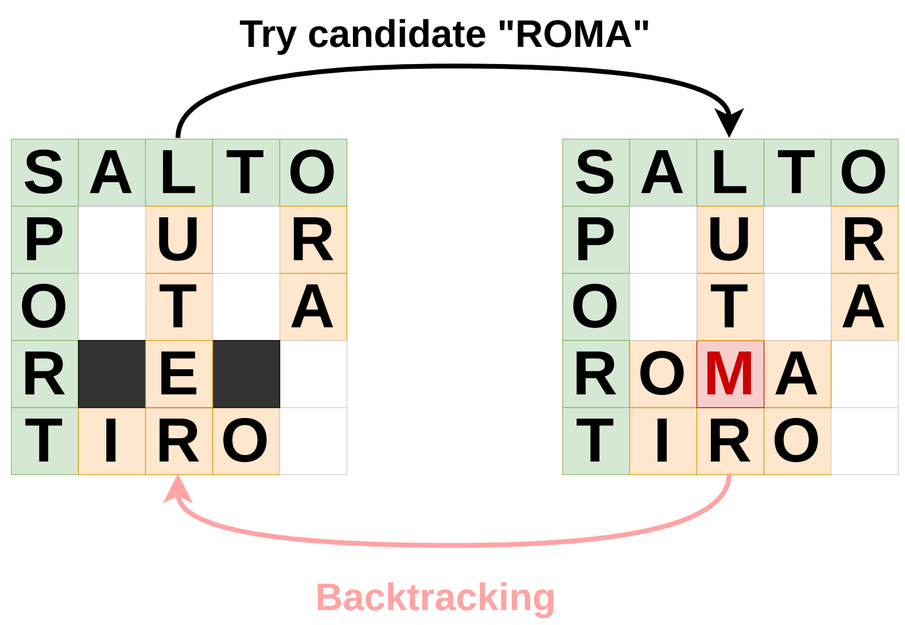
Methodology: Special Length Tokens

- Model: *it5-efficient-small-el32*
- **Challenge:** Standard models struggle with exact character counting
- **Solution:** Dual-constraint mechanism with special anchor tokens

Methodology: CSP with Backtracking

- **Two-stage solving:**
  1. *Seeding:* fix high-confidence predictions as anchors
  2. *DFS Backtracking:* fill slots greedily, backtrack on conflicts





Results

Model	Epochs	Acc@1	Acc@10	MRR
Dict + tokens	23	**0.58**	**0.75**	**0.63**
Dict + tokens	22	0.57	0.75	0.62
No tokens	23	0.38	0.54	0.43
Baseline	-	0.40	0.62	0.46

Results

Algorithm	Cand.	Char Acc.	Word Acc.	Grid Acc.
22ep + Dict	600	0.92	**0.85**	**0.34**
22ep, No Dict	600	0.92	0.85	0.32
23ep + Dict	600	**0.93**	0.84	0.28
Baseline	-	0.73	0.58	0.08

Conclusions

- Special length tokens effectively encode constraints into T5 generation.
- CSP solver with backtracking successfully fills crossword grids.
- Neural + CSP outperforms pure IR baselines.

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